CHAPTER 5

COORDINATION COMPOUNDS

CHAPTER ESSENCE AT A GLANCE:
Werner's coordination theory, IUPAC naming of complex compounds, isomerism, crystal field splitting in octahedral and tetrahedral complexes, and color.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **Werner's Coordination Theory:** Central metal atoms possess two distinct valencies: primary valency which is ionisable and corresponds to oxidation state, and secondary valency which is non-ionisable, directional, and dictates coordination number and spatial geometry.

• **Isomerism in Coordination Entities:** Coordination compounds exhibit structural isomerisms including ionisation, linkage, coordination, and solvate forms, as well as stereoisomerism comprising cis-trans geometrical alignments and non-superimposable chiral optical enantiomers.

• **Crystal Field Theory & Orbital Splitting:** Electrostatic ligand fields remove the degeneracy of metal d-orbitals, splitting them into lower t2g and higher eg sets in octahedral fields, and inverted e and t2 sets in tetrahedral complexes.

• **Spectrochemical Series & Spin States:** Ligands are organized by their relative field splitting strength; strong-field ligands overcome electron pairing energy to form low-spin diamagnetic or weakly paramagnetic complexes, whereas weak-field ligands yield high-spin configurations.

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\mathrm{CFSE}_{\mathrm{oct}} = (-0.4\, n_{t_{2g}} + 0.6\, n_{e_g})\Delta_o + mP$$

Crystal Field Stabilization Energy (CFSE) in octahedral complexes

$$\Delta_t = \frac{4}{9}\Delta_o$$

Relationship between tetrahedral and octahedral crystal field splitting energies

$$\beta_4 = \frac{[\mathrm{ML_4}]}{[\mathrm{M}][\mathrm{L}]^4} \quad \text{and} \quad \log \beta_4 = \sum_{i=1}^{4} \log K_i$$

Overall stability constant of coordination complexes in solution

$$\mathrm{I^- < Br^- < Cl^- < F^- < OH^- < H_2O < NH_3 < en < CN^- < CO}$$

Spectrochemical series in increasing crystal field splitting order

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** When the coordination sphere is ANIONIC (e.g. $[Fe(CN)_6]^{4-}$), the central metal is called FERRATE, not iron!

• **Alert:** [Valuation Rubric] Step 1 [Oxidation State & Configuration]: Find oxidation state of metal and write valence d-configuration .

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. State the main postulates of Werner's coordination theory. Differentiate primary and secondary valency. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Werner's Postulates]** 1. Metals exhibit two types of valency: Primary and Secondary. 2. Primary valency: Ionisable, satisfied by negative ions, corresponds to oxidation state. 3. Secondary valency: Non-ionisable, satisfied by neutral molecules or negative ions, corresponds to coordination number. 4. Secondary valencies have fixed spatial orientations in space giving definite geometry (octahedral, tetrahedral, square planar). [3 Marks]

• **Key Exam Result:** Primary (ionisable, OS) vs Secondary (non-ionisable, CN, directional geometry)

2. Define: (a) Coordination sphere, (b) Coordination number, (c) Chelating ligand, (d) Ambidentate ligand. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Definitions]** (a) Coordination Sphere: Central metal atom or ion and attached ligands enclosed in square brackets [M L_n]. Non-ionisable. (b) Coordination Number: Total number of coordinate (dative) bonds formed by ligands with the central metal ion. (c) Chelating Ligand: Di- or polydentate ligand that binds through two or more donor atoms simultaneously to form a stable 5- or 6-membered cyclic ring with metal (e.g. oxalate, ethylenediamine). [3 Marks]

• **Key Exam Result:** Definitions: Sphere [ML_n], CN (total dative bonds), Chelating ligand (ring former)

3. Write IUPAC names for: (a) [Co(NH3)5Cl]Cl2, (b) K3[Fe(CN)6], (c) [Pt(NH3)2Cl(NO2)], (d) [Cr(en)3]Cl3. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: IUPAC Names]** (a) $[Co(NH_3)_5Cl]Cl_2$: Pentaamminechloridocobalt(III) chloride. (b) $K_3[Fe(CN)_6]$: Potassium hexacyanidoferrate(III). (c) $[Pt(NH_3)_2Cl(NO_2)]$: Diamminechloridonitrito-N-platinum(II). [3 Marks]

• **Key Exam Result:** (a) Pentaamminechloridocobalt(III) chloride | (b) Potassium hexacyanidoferrate(III) | (c) Diamminechloridonitrito-N-platinum(II)

4. Explain structural isomerism in coordination complexes: Ionisation, Linkage, Coordination, and Hydrate isomerism. [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Four Structural Isomerisms]** 1. Ionisation Isomerism: Exchange of ions between coordination sphere and ionization sphere (e.g. $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$). 2. Linkage Isomerism: Ambidentate ligand coordinates through different donor atoms (e.g. $[Co(NH_3)_5(NO_2)]Cl_2$ (nitro, N-donor) and $[Co(NH_3)_5(ONO)]Cl_2$ (nitrito, O-donor)). 3. Coordination Isomerism: Interchange of ligands between cationic and anionic complex entities (e.g. $[Co(NH_3)_6][Cr(CN)_6]$ and $[Cr(NH_3)_6][Co(CN)_6]$). 4. Solvate/Hydrate Isomerism: Water molecules differ as ligand vs crystallization water (e.g. $[Cr(H_2O)_6]Cl_3$ (violet) and $[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O$ (grey-green)). [4 Marks]

• **Key Exam Result:** Ionisation, Linkage, Coordination, Solvate isomerisms

5. Explain Crystal Field Splitting in an octahedral coordination entity (t2g and eg splitting diagram). [Score: 4 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Octahedral Field Splitting]** Six ligands approach along x, y, z axes. Orbitals along axes (d $\{x^2-y^2\}$, d $\{z^2\}$) experience greater electrostatic repulsion and rise in energy by +0.6 Δ_0 (e g set). Orbitals between axes (d {xy}, d {yz}, d {zx}) experience less repulsion and lower in energy by -0.4 Δ_0 (t {2g} set). [3 Marks]

• **Key Exam Result:** Splitting energy Δ_0: e g (+0.6 Δ_0) and t {2g} (-0.4 Δ_0)

6. What is Spectrochemical series? Differentiate between strong field and weak field ligands. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Spectrochemical Series]** Arrangement of ligands in increasing order of crystal field splitting energy (Δ_0): I^- < Br^- < S^{2-} < SCN^- < Cl^- < N_3^- < F^- < OH^- < $C_2O_4^{2-}$ < H_2O < NCS^- < $edta^{4-}$ < NH_3 < en < CN^- < CO. [2 Marks]

• **[Step 2: Weak vs Strong Field]** Weak field ligands (e.g. halides, H_2O) produce small Δ_0 (Δ_0 < P); high spin complexes. Strong field ligands (e.g. CN^-, CO, NH_3) produce large Δ_0 (Δ_0 > P); electrons pair up, forming low spin complexes. [1 Mark]

• **Key Exam Result:** Strong field (Δ_0 > P, low spin) vs Weak field (Δ_0 < P, high spin)

7. Why are tetrahedral complexes always high spin complexes? [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Explanation]** In tetrahedral complexes, crystal field splitting energy Δ t is always smaller than pairing energy P because: 1. Only 4 ligands instead of 6. 2. None of the d-orbitals point directly at ligands. Since Δ t = (4 divided by 9) Δ_0 < P, pairing of electrons is energetically unfavourable; electrons always enter higher t_2 orbitals before pairing, forming high spin complexes exclusively. [2 Marks]

• **Key Exam Result:** Δ t < P always; pairing energy cannot be overcome